

# Parallel Processing in memory

The memory architecture in parallel computers are classified in 3 categories

## 1. Shared memory architecture:

It is used in multiprocessors.

Shares a common memory and only one processor can access the memory at a time.

Shared memory multiprocessors can be divided into two classes based on memory access time.

### 1. UMA

### 2. NUMA (Non-Uniform Memory Access)

### 3. COMA (cache only memory architecture) it is a special case of NUMA.

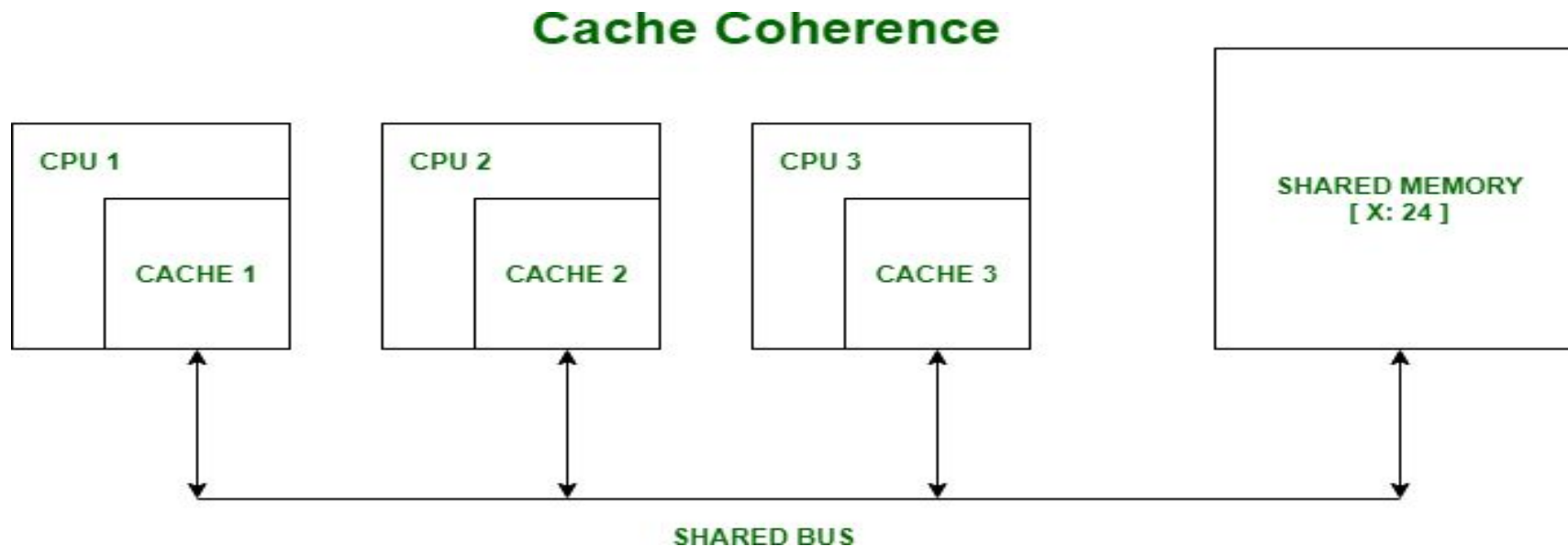
# UMA

## Model

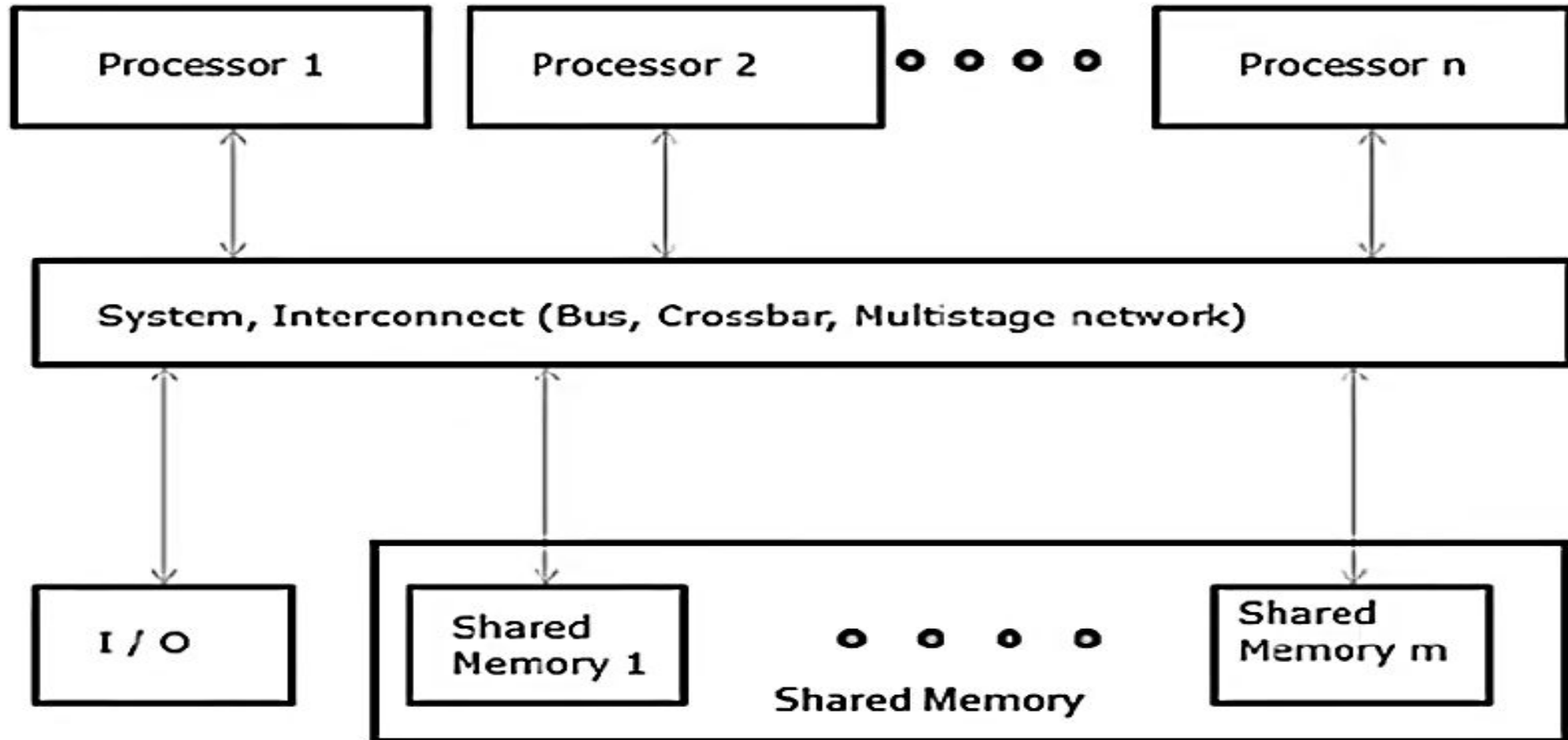
- In this model the physical memory is shared uniformly by all processors.
- All the processors have equal access time to all the memory location.
- It is used in **symmetric multiprocessor (SMP) machines**.
- Sometimes each processor may have its own private cache also known as cc-UMA (cache coherent-UMA)

- **Cache coherence** : In a multiprocessor system, data inconsistency may occur among adjacent levels or within the same level of the memory hierarchy. In a shared memory multiprocessor with a separate cache memory for each processor, it is possible to have many copies of any one instruction operand: one copy in the main memory and one in each cache memory.

When one copy of an operand is changed, the other copies of the operand must be changed also. Example : Cache and the main memory may have inconsistent copies of the same object.



# UMA multiprocessor model



# NUMA Model

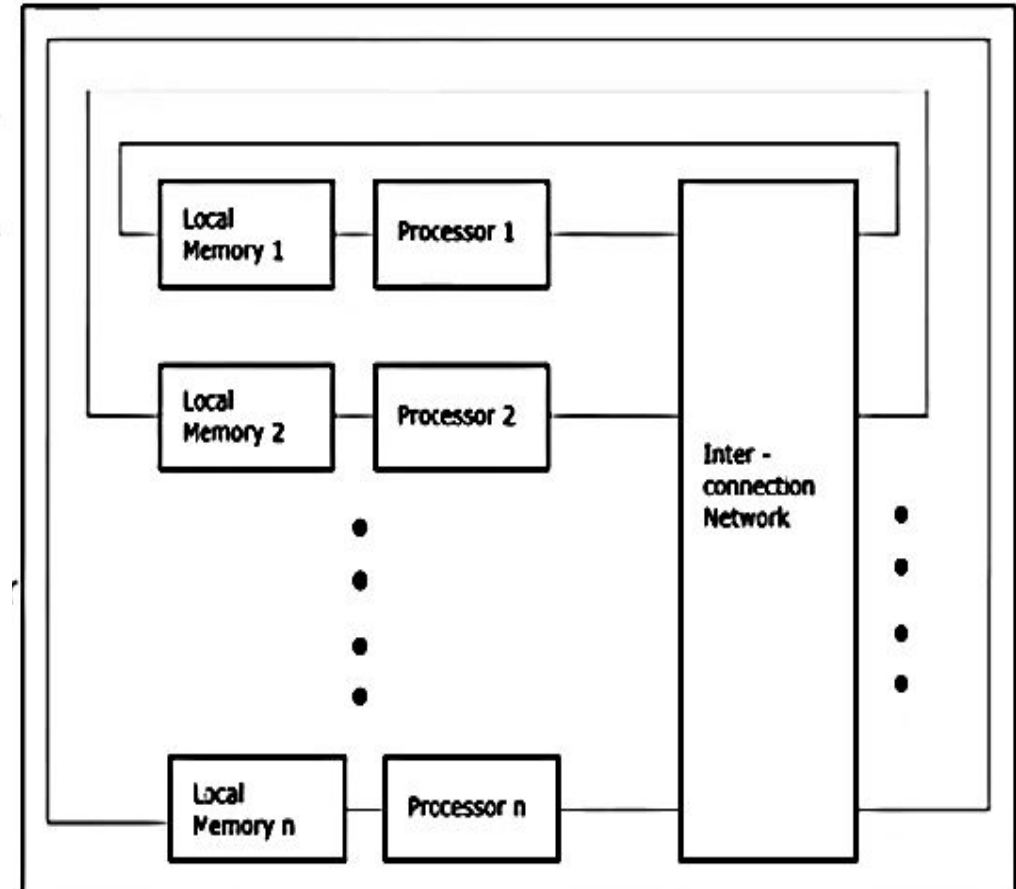
A **NUMA Multiprocessor model** ~~shared-memory~~ **system** in which the **access time varies with the location of the memory word**

The **shared memory** is physically **distributed** among all the processors, called **local memories**. The collection of all local memories forms a **global address space** which can be accessed by all the processors.

It is **faster to access a local memory** with a local processor

The **access of remote memory** attached to other processors is **slower** due to the added **delay** through the interconnection network

- Sharing remote memory means sharing of local memory of other processors.
- NUMA can be used to design multiple SMPS.
- In case the local computer has its own cache and it is maintaining the cache coherence then it is called CC- NUMA.

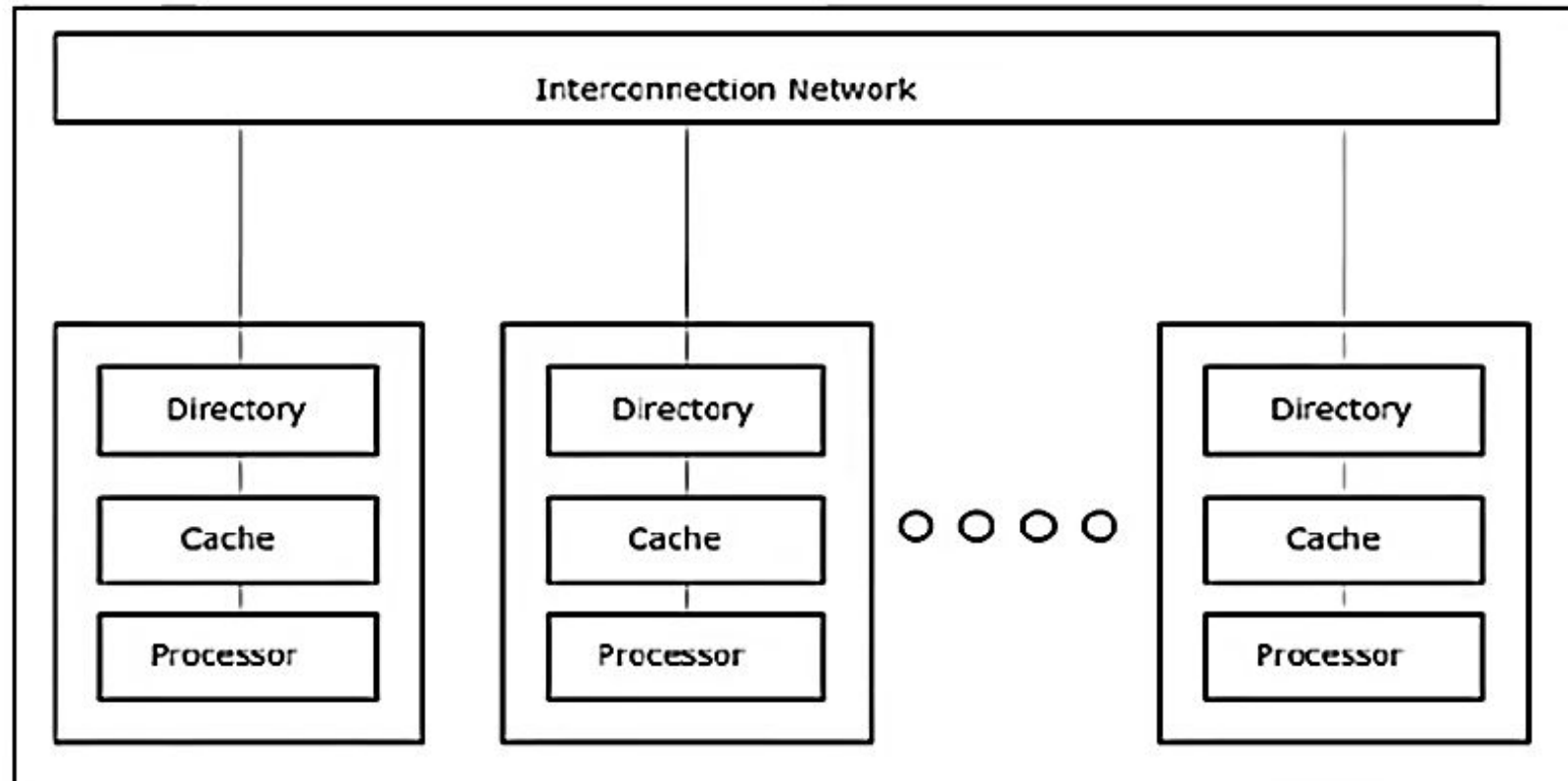


# COMA

The **COMA model** is a *special case of NUMA machine* in which the distributed main memories are converted to **Caches**

All caches form a **global address space** and there is no memory hierarchy at each processor node

**Remote cache access** is assisted by the distributed **cache directories**



# Distributed Memory Architecture

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Distributed memory systems consists of multiple computers, often called *nodes*, interconnected by a *message-passing network*. Each node is an autonomous computer consisting of a processor, local memory and sometimes attached disk or I/O peripheral

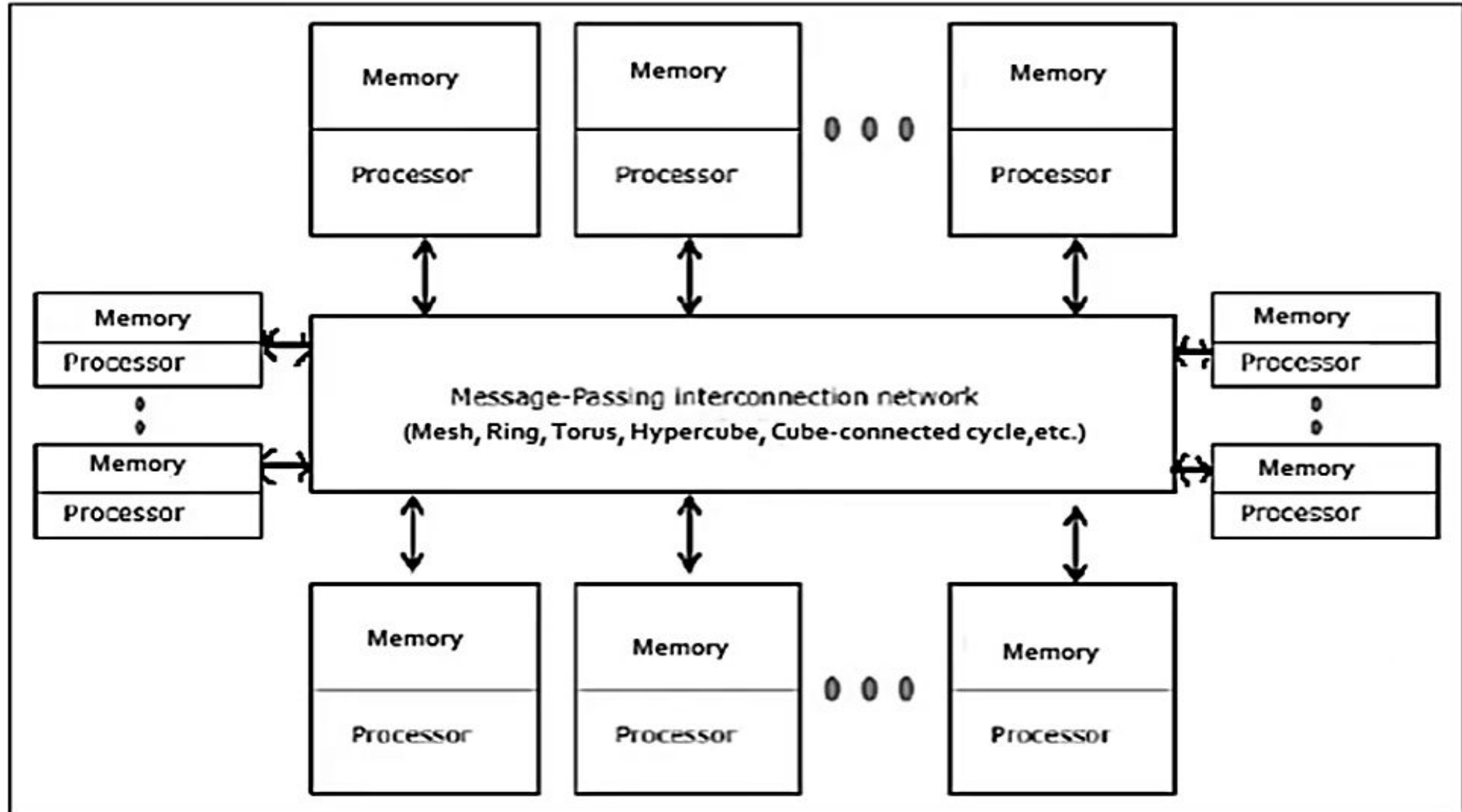
Distributed Memory are **Loosely coupled systems** as processors have their own local memory

Processors have their own local memory. Memory addresses in      processor do not map to another processor, so there is **no concept of global address space** across all processor.

Because each **processor** has its **own local memory**, it operates **independently**. Changes it makes to its local memory have no effect on the memory of other processors. Hence, the **concept of cache coherency does not apply**.

When a processor needs access to data in another processor, it is done by passing message between processors, which is usually the task of the *programmer* to explicitly define how and when data is communicated.

# Distributed Memory System





# Hybrid Memory Architecture

The largest and fastest computers in the world today employ **both shared and distributed memory architectures**.

The **shared memory component** is usually a cache coherent SMP machine. Processors on a given SMP can address that machine's memory as global.

The **distributed memory component** is the networking of multiple SMPs. SMPs know only about their own memory - not the memory on another SMP. Therefore, network communications are required to move data from one SMP to another.

Current trends seem to indicate that this type of memory architecture will continue to prevail and increase at the high end of computing for the foreseeable future.

*Advantaged and Disadvantages:* Whatever is common to both shared and distributed memory architectures

